

Willamette Applicant Trajectory: Predicting Student College Enrollment Destination

Data 510 Capstone Project Write-up

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2026-08-02

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1 Title, Team, and Affiliations

Project title:

Willamette Applicant Trajectory: Predicting Student College Enrollment Destination

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Repository: <https://github.com/Data-Science-Capstone-Project-2026/Capstone2026-Applicant-Trajectory>

2 Abstract

Willamette University's Admissions Office and enrollment management team invest significant resources in recruiting prospective students, yet a meaningful portion of admitted applicants ultimately choose to enroll elsewhere. To better allocate resources, past enrollment data was analyzed using Multiple Linear Regression and XGBoost to identify which student features can predict what category of school a student will enroll in. It was found that an applicant's region of origin has the most impact on where they will enroll, as well as their academic strength and financial situation. Enrollment in Northwest 5 schools specifically is also greatly impacted by visitation and legacy status. These findings are limited by the smaller sample size and time period of data available, limiting the generalizability to other institutions. With these findings, the Willamette University Admissions and Enrollment teams can optimize their time and resources to recruit new students.

3 Introduction: Problem and Research Questions

Problem statement:

Willamette University's Admissions Office and enrollment management team invest significant resources in recruiting prospective students, yet a meaningful portion of admitted applicants ultimately choose to enroll elsewhere.

Currently, the Admissions office lacks a data-driven framework for segmenting admitted applicants by their predicted matriculation destination, which means that recruitment and yield strategies must be applied broadly rather than targeted to what differentiates Willamette from the other schools an applicant may be looking at. This limits the effectiveness of outreach efforts at a critical stage in the enrollment funnel. Understanding which type of institution a student is likely to choose would allow the team to tailor materials, financial aid messaging, and communications to what is most likely to influence a student's decision.

Our project aims to address that gap by identifying applicant features of importance through exploratory data analysis and clustering, ultimately building a predictive model trained on the past 3 years of historical applicant data matched to National Student Clearinghouse enrollment data. We analyze two destination groupings as part of an iterative refinement process. An initial grouping (Version 1) classified admitted students by destination type using a broader geographic scope and including campus engagement history as predictive features. Following stakeholder review, a second grouping (Version 2) narrowed the geographic scope and restricted features to application-inherent applicant characteristics, better isolating the factors Admissions can act on at the point of application. Both models classified admitted students into one of four primary destination categories, with a residual Other category in the first version: **NW5** (Willamette University, Reed College, Lewis & Clark College, Whitman College, and University of Puget Sound), **Three Large Public** (Oregon State University, University of Oregon, and Portland State University), **Local Public**, and **Local Private**, with the Version 1 and Version 2 groupings differing in how the Local categories and Other are constructed (detailed in Section 6). Version 2 is emphasized in results as the primary deliverable given its closer alignment with stakeholder priorities and its exclusive presentation in the accompanying poster. Predictions generated by this model will be used by the Admissions Office to segment incoming applicants and enable targeted outreach strategies accordingly.

Research Questions: 1. What applicant characteristics most strongly predict whether an admitted student will enroll at a private NW5 institution or a large public university? 2. Are there additional meaningful enrollment destinations beyond the initial categories that arise from the data, and what applicant profiles define those groups?

Changes since M1 proposal:

Delta 1 - Two-stage destination categorization: The initial framing considered in M1 (a private NW5 vs. large public binary, with potential additional segments to be determined) was implemented in two stages. Version 1 established four categories plus Other, using a broad geographic scope for the Local groupings. Version 2 refined this by narrowing the geographic scope and adjusting how Local Public and Local Private are constructed. Both versions are reported; rationale for the refinement is detailed in Section 6.

Delta 2 - Geographic scope comparison: Version 1 retained enrollment destinations nationally, including California institutions within the Local groupings. Version 2 narrowed this scope in response to stakeholder feedback that California-heavy signal was likely skewing results away

from the Pacific Northwest institutions most relevant to Willamette’s competitive set. Both scopes are analyzed for comparison; rationale is detailed in Section 6.

Delta 3 - Feature scope comparison: Version 1 included campus visit and engagement history variables as predictive features. Version 2 restricted features to application-inherent applicant characteristics, per stakeholder interest in signals available at the point of application. Both feature sets are analyzed for comparison; rationale is detailed in Section 8 (Methods and Evaluation).

4 Impact and Stakeholders

Who Benefits

Primary Stakeholders

The primary beneficiaries of this project are William Mullen, VP for Enrollment Management, and the broader Admissions team at Willamette University. If the model performs well, the Admissions office gains a practical tool that can assist with a more efficient allocation of recruitment and outreach resources and, ideally, improved yield rates over time.

Indirect Stakeholders

The Willamette Marketing team stands as an indirect beneficiary, as the model insights could inform messaging and campaign strategies. Prospective students could also benefit from a more targeted outreach that contains relevant communication to their needs.

Organizational Impact

A reliable prediction model would give the Willamette Admissions office a replicable, data-informed process that can be refined for each cycle and utilized in yield strategy. Over time, even modest improvements in the enrollment rate can have meaningful financial and institutional implications.

5 Related Work and Background

Across the country, higher education institutions are preparing for or beginning to feel the effects of the enrollment cliff- the idea that after 2025, colleges and universities will begin to see a dramatic decline in enrollment due to changing demographics [1]. Intentional actions are needed to ensure the longevity of these institutions.

The need for intentional actions has driven others to conduct research and look for solutions. Predicting whether students will enroll, and where they will enroll, has been a topic of research in the past [3]. This project specifically looks at whether a student will enroll at California State University Channel Islands or another institution using information found from applicants to

the University. Researchers at the University of New Mexico (UNM) conducted similar research into the destination of students who applied. [2] This project found that State Financial awards, applying after February, and having a 3.0-3.5 GPA were some of the most predictive features of enrollment at UNM. Both of these examples were done at public universities and were predicting whether students would enroll or not. This project addresses the gap in research by predicting destination type as well as using data from a private, liberal arts university, potentially enabling more targeted, comparison-aware outreach.

6 Data and Data Engineering

Sources:

Dataset	Source	Row Counts	Date Range	Refresh Policy	Repo Location
Slate applicant data	Technolutions Slate CRM (SFTP)	9404	2023-2025	Historical pull covering a 3-year window. Frozen.	data/raw/willamette_app_details.csv
NSC enrollment 2023	National Student Clearinghouse (CSV)	2,353	2023	Historical pull. Frozen.	data/raw/nsc_fall23.xlsx
NSC enrollment 2024	National Student Clearinghouse (CSV)	2,120	2024	Historical pull. Frozen.	data/raw/nsc_fall24.xlsx
NSC enrollment 2025	National Student Clearinghouse (CSV)	2,125	2025	Historical pull. Frozen.	data/raw/nsc_fall25.xlsx
Carnegie Classification of Institutions of Higher Education	Website URL	3928	2025	Manual pull. Static reference data. Frozen.	data/external/carnegie_classification.csv

Engineering summary:

Repository structure:

- data/raw/: Original files as received from the SFTP server and Carnegie Classification website. Gitignored. Never modified.
- data/external/: Supplementary reference data not generated by the team. The Carnegie Classification of Institutions of Higher Education file, used to categorize enrollment destinations by institution size and public/private status.
- data/interim/: Outputs from cleaning, joins, and intermediate transformations, staged prior to final processing.
- data/processed/: Analysis-ready dataset used for visualization and modeling.
- src/Capstone_data_cleaning/: SQL ingestion, cleaning, schema creation, and migration scripts.
- src/EDA/ - Generate figures and reporting visualizations for exploratory data analysis.
- src/features/: Clean, join, and build features from data/raw/ into data/processed/. Scripts to create target variables and feature engineering for models exist in this folder.
- src/models/: Train, evaluate, and run inference on processed data. Files are named as model type and an additional _v2 for version 2 models.

Pipeline overview

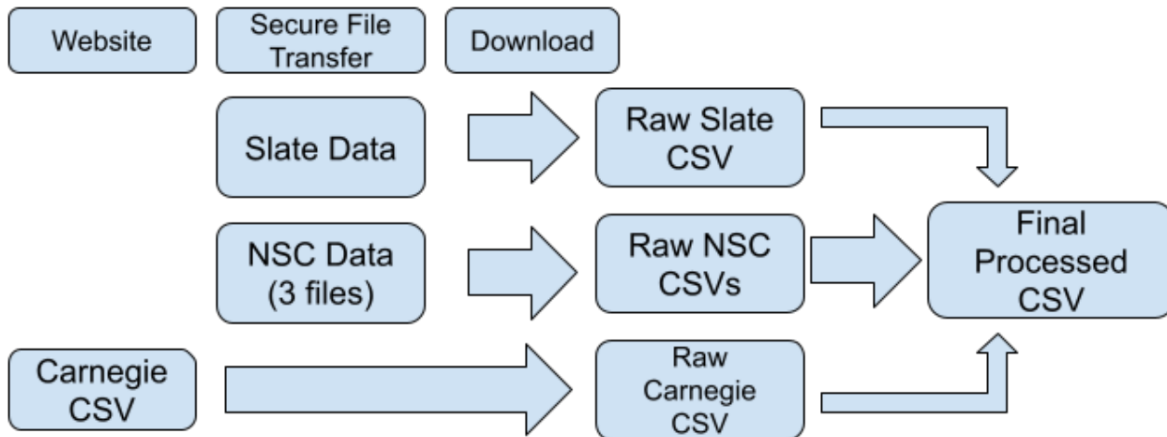


Figure 1: Pipeline

The pipeline proceeded in the following stages:

- **Ingestion:** Raw Slate and NSC exports, and the Carnegie reference file, were pulled into data/raw/ as a one-time historical pull, consistent with the project's frozen 3-year data scope.

- **Staging cleanup:** The NSC tables were standardized to a common schema and unioned into a single staging table, with duplicate applicant records resolved by retaining each student's earliest recorded enrollment term. The Slate applicant staging table was cleaned and duplicate applicant records resolved by retaining the earliest submitted application per applicant.
- **Schema creation and migration:** A normalized capstone schema was created to hold the cleaned data (see ERD below), and cleaned staging data was migrated into it.
- **Flat file creation:** The normalized tables were joined into a single analysis-ready flat file, matching Slate applicants to their NSC-confirmed enrollment destination, for use in exploratory analysis and modeling.

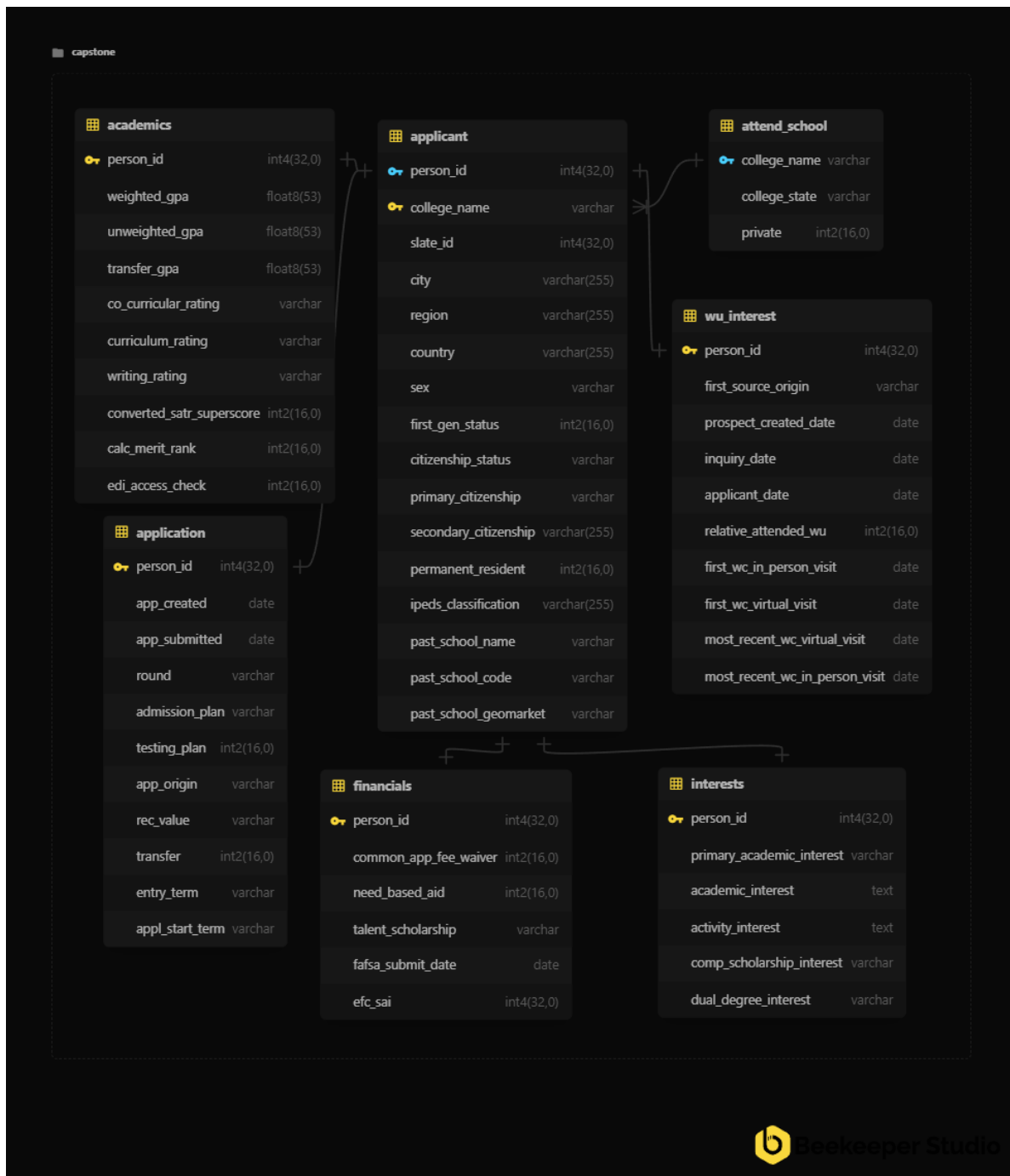


Figure 2: ERD

Destination Category Construction This project has two versions of processed datasets,

with modeling and XGBoost specific datasets for each version. The construction of the second version reflects the iterative refinement described in the Introduction.

Version 1 used a “Local Large Public” category (public institutions above 20,000 students, outside the Three Large Public group) and defined Local Large Public and Local Private as universities in Oregon, Washington, and California. The Other category held the other enrollment destinations nationally.

Version 2 narrowed the geographic scope of the Local categories to Oregon and Washington institutions only after stakeholder review. The Local Large Public category was also replaced with a broader Local Public category (all Oregon/Washington public institutions outside the Three Large Public group, regardless of size), since the size-restricted version yielded an impractically small sample. The Other category was dropped for this version of the analysis, since our stakeholder wanted to focus on the main target audience, which are Oregon and Washington high school students.

7 Ethics and Responsible Use

Privacy:

None of the data used in this project contains Personally Identifiable Information (PII), protecting the identities of the students represented in the dataset. All PII was removed prior to data landing in data/raw/, ensuring that no personally identifiable information is present at any stage of the repository. In the original raw data, birthdate, sex, and gender were present together, which the team identified as a combination that could reasonably enable re-identification of a student even in the absence of direct identifiers; the query co-developed within Slate was updated to remove these columns before re-export to the SFTP server.

Both project owners signed a formal confidentiality agreement governing the use of this data. Access to the source data is restricted to credentialed team members connected to the Willamette campus network, using a combination of a raw password, a public/private key pair held exclusively on the two team members’ local devices, and an IP whitelist. The data has not been and will not be shared with other members of the capstone cohort, external stakeholders beyond the primary stakeholder, or the public. No specific use constraints were imposed by either Technolutions or the National Student Clearinghouse beyond the confidentiality agreement, and the data is used solely for the purposes described in the project proposal, with no repurposing outside this project’s scope.

Fairness / equity:

The primary fairness consideration in this project concerns the use of predictive models trained on data that has been unavoidably shaped by existing inequities in collegiate recruitment and enrollment. Applicant characteristics such as financial aid status, recommender ratings, and geographic origin can correlate with socioeconomic and demographic factors outside a student’s

control. Model outputs are closely evaluated for whether they disproportionately disadvantage any applicant subgroup in how outreach is targeted, and are subject to revision if they are found to reinforce rather than reduce existing inequities in access and yield.

Limitations of setting:

This analysis is scoped to a single institution and the destination categories reflect the Willamette’s specific competitive set and stakeholder priorities rather than general higher education destinations. Findings should not be assumed to generalize to other institutions’ applicant pools or to a different regional context.

Non-causal framing:

Applicant characteristics identified as predictive of a given destination category should be interpreted as associative signals useful for segmentation, not as causal drivers of a student’s enrollment decision. The models are designed to support targeted outreach based on predicted likelihood, not to explain or claim why a student chooses one institution type over another.

8 Methods and Evaluation

Design: This project started with an exploratory stage to construct and validate features and applicant groupings, followed by a supervised classification stage to predict enrollment destination. Both stages were run under Version 1 and Version 2 groupings that are described in Section 6 Data and Data Engineering.

Primary analysis: Exploratory data analysis was conducted in R and done across the different feature groupings as shown in the ERD schema in Section 6. Applicant attributes were then transformed into model-ready features (feature engineering).

Clustering was applied to identify natural groupings in enrollment destination patterns. To assess whether the cluster membership aligned meaningfully with the enrollment destination categories, a chi-square test of independence was run with Cramer’s V used to quantify the strength of the association.

Three classification approaches were compared for the predictive task: Multinomial Logistic Regression (MLR) as an interpretable baseline, Support Vector Machines (SVM) as an alternative classifier, and XGBoost, selected for its native handling of missing values. The highest performing model was then focused on for hyperparameter tuning.

Evaluation metrics: Model performance was evaluated using accuracy and macro F1 score, with macro F1 given particular weight to account for class imbalance across destination categories. Performance is benchmarked against a majority-class baseline (always predicting the most common destination category).

Train/Test Split: Models were evaluated using an 80/20 train-test split to ensure that there is adequate data to validate against, with no separate validation set held out.

9 Results and Visualizations

Exploratory Data Analysis:

Note: The following results are presented under the Version 2 destination grouping. Both the Version 1 and Version 2 groupings were analyzed at this stage and can be found in the repository, but the underlying trends were consistent across both. Group-level values shifted modestly, but the direction and interpretation of each result did not change. Version 2 is shown here as the primary reference, consistent with its emphasis throughout this write-up.

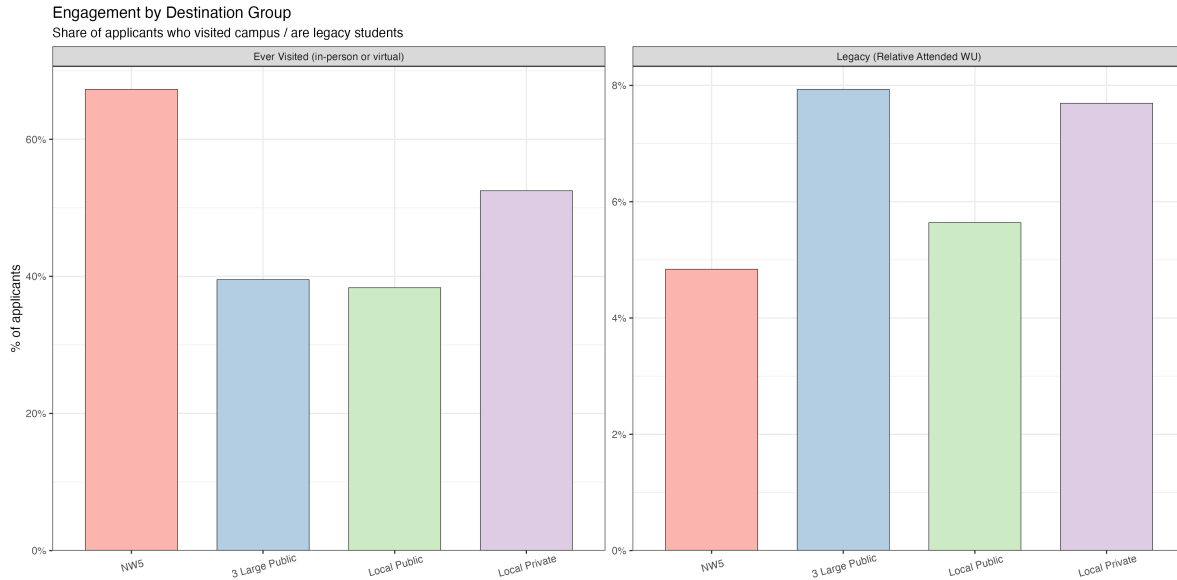


Figure 3: Figure 1: Engagement by Destination Category

Students who ultimately enroll at an NW5 institution are more likely to have visited campus, either physically or virtually, which is consistent with the geographic clustering of NW5 institutions in the Pacific Northwest. A notable exception is legacy status: applicants with a parent or relative who attended Willamette show the highest rate of enrolling at one of the three large public universities (OSU, UO, or PSU) rather than at Willamette or another NW5 school. One plausible interpretation is that these students wanted to remain near family or in a familiar state, but sought a substantially different campus experience, deciding on a large public university rather than a small private one.

Financial need signals vary meaningfully by destination group. The EFC/SAI distribution shows that students enrolling at the three large public universities generally come from lower-income households than students in other destination groups, and this group also requested the highest proportion of application fee waivers. FAFSA submission rates are consistently high across all destination groups, but need-based aid consideration is highest among Local

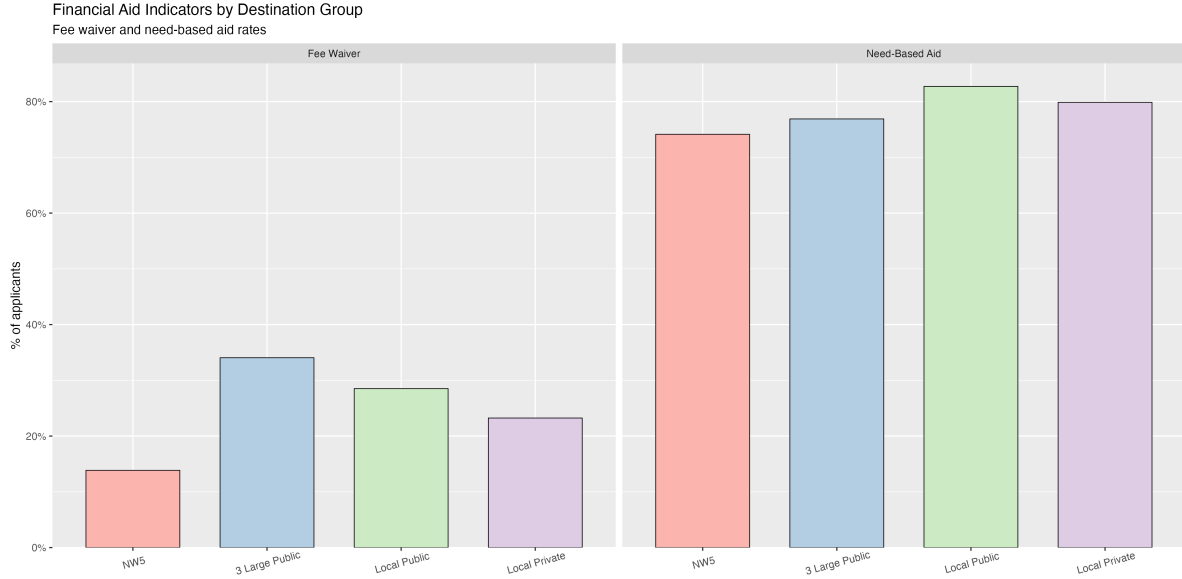


Figure 4: Figure 2: Financial Aid Indicators by Destination Category

Public enrollees, with Three Large Public close behind. Across all three financial indicators, NW5 enrollees show the lowest financial need proportionally.

Clustering and Category Validation

Applicant profiles were clustered ($k = 4$) using the engineered feature set, producing groups of moderately even size (ranging from approximately 244 to 973 applicants). The heatmap displays the five features found most influential in differentiating clusters: weighted GPA, EFC/SAI, days from prospect to application, recommender value, and curriculum rating.

Group 2 and Group 3 emerge as the strongest opposites. Group 2 shows the strongest academic profile across the board (GPA, curriculum rating, writing rating) along with the highest calculated merit rank and a larger share of students opting into standardized testing; this group also takes the longest, on average, to submit their application. Group 3 shows the opposite academic pattern: lower ratings across academic indicators, lower merit ranks, and faster application turnaround, despite comparable financial need to Group 2. Group 1 stands out primarily on financial profile rather than academics: it ranks in the middle for academic strength and merit rank, but shows lower EFC/SAI alongside higher rates of fee waiver requests, need-based aid consideration, and FAFSA submission, along with a high proportion of first-generation students.

Two complementary crosstabulations were examined: one showing how each cluster's members distribute across destination categories, and one showing how each destination category's students distribute across clusters. Destination composition is fairly even across clusters overall, with a few notable exceptions. Cluster 1 holds the highest share of Three Large Public enrollees

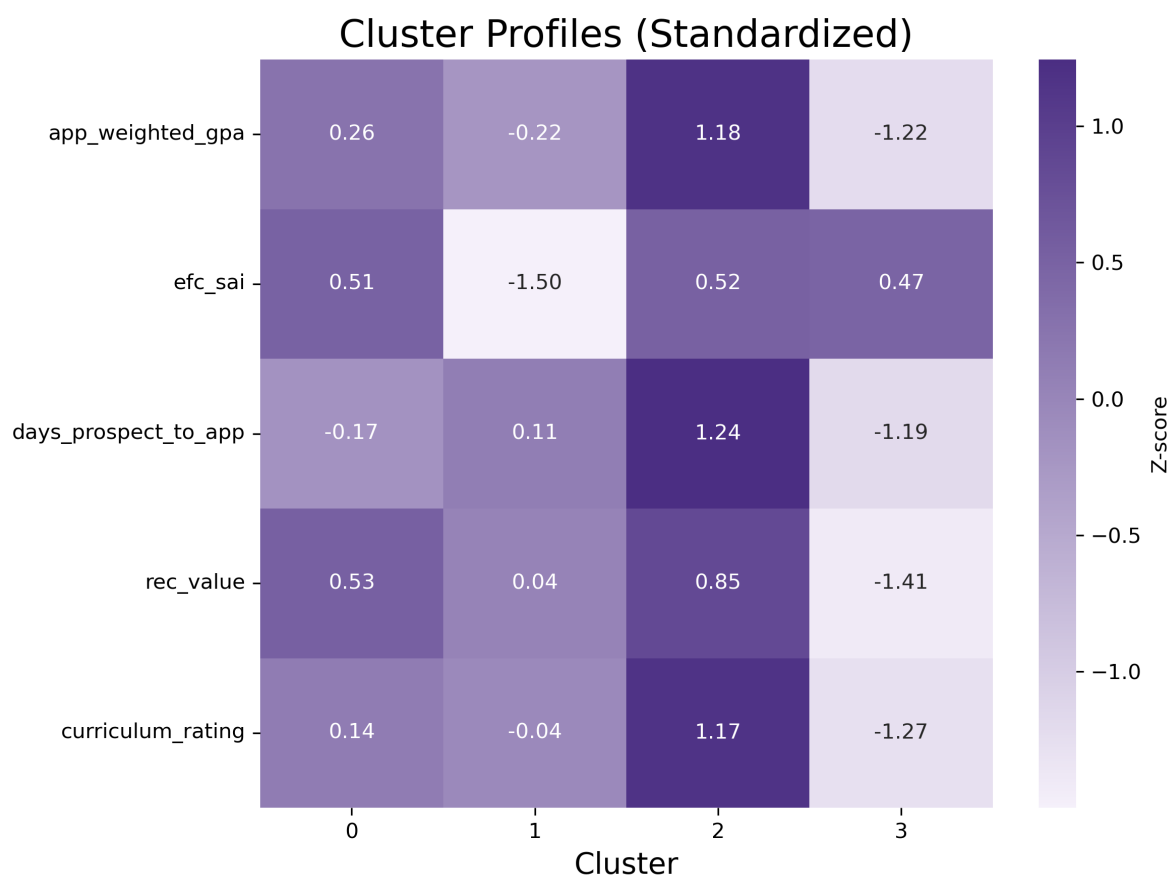


Figure 5: Figure 3: Cluster Profiles (Standardized)



Figure 6: Figure 4: Cluster Membership by Destination Category

and the lowest share of NW5 enrollees. Cluster 3, characterized by lower academic ratings, holds the highest share of Local Private enrollees and the lowest share of NW5 enrollees; however, Cluster 2, characterized by high academic ratings, also holds the largest share of Local Private enrollees alongside the largest share of NW5 enrollees, indicating that strong academic profiles are not exclusive to NW5 students. Viewed from the destination side, NW5 enrollees are most commonly drawn from Cluster 2, but Cluster 0, a group with solid academics and average financial need, makes up an even larger proportion of NW5 enrollees and is in fact the most common cluster across nearly every destination group.

Taken together, the clusters represent coherent, interpretable combinations of academic strength, financial need, and application timeline behavior. However, these combinations are not strongly predictive of enrollment destination on their own, at least at $k = 4$ with the current feature set.

Additionally, a chi-square test of independence between cluster membership and destination category returned a p-value below 0.05, leading to rejection of the null hypothesis: cluster membership and destination category are not statistically independent. However, Cramer's V (which quantifies the strength of association between two categorical variables on a 0–1 scale) was extremely low at 0.076, indicating that while the association is not due to chance, it is weak in practical terms. This directly addresses Research Question 2: while the clusters that emerge from the data are internally coherent and interpretable, they do not meaningfully

validate or refine the destination categories, and **cluster membership alone is not a strong standalone predictor of enrollment destination.**

Classification Approaches As the baseline model, class-weighted Multinomial Logistic Regression achieved 43% overall accuracy (macro F1 = 0.43), with precision and recall varying meaningfully by destination category. Recall was strongest for Local Public (0.58) and Three Large Public (0.56), and weakest for Local Private (0.25), indicating the baseline struggles most to correctly identify Local Private enrollees despite reasonable precision (0.50) when it does.

The next model used was the Support Vector Machine, which resulted in an overall accuracy of 45%. The strongest recall was for Local Private Schools (0.60), with the weakest recall for Northwest 5 Schools (0.14), indicating that the SVM struggles most to correctly identify Northwest 5 enrollees despite reasonable precision (0.52) when it does.

The final XGBoost Model has an overall accuracy of 43%, with the highest precision of 50% in the Northwest 5 category and the highest recall of 52% in the Local Private and Three Large Public categories.

The model found that overall, the most important features influencing a student’s destination category are their region of origin, whether or not they were a student athlete, and the number of days it took for them to submit their application after they started it. However, these results vary across destination category. For all 4 categories, region of origin was the most influential factor. However, being a student athlete was only influential for local private schools or the three large public schools, with student athletes more likely to go to local private schools and less likely to go to one of the three large public schools, as seen in figures A2 and A3. Academic categories such as the converted SAT/ACT superscore, the applicant’s weighted GPA, and the rating of high school curriculum difficulty were more influential in the private school categories (NW5 and Local Private) than for the public schools.

The four most influential features for each destination category are listed below.

Destination Category	Most Influential Feature	Second most influential feature	Third Most Influential Feature	Fourth Most Influential Feature (if needed)
Northwest 5	Region	SATR Superscore	First Generation Status	Days to Application Submission
Local Private	Region	Student Athlete Status	Weighted GPA	Curriculum Rating
Three Large Public	Region	Student Athlete Status	EFC/SAI	Not needed
Local Public	Region	Days to Application Submission	Admission Plan	Not needed

The directionality of the feature importance was found to be varied for each of the destination categories as well. The strength of these relationships was strongest for the applicant's region of origin for all categories. The next strongest impact on the model output varied from an absolute value of 0.05 to 0.75 (on a scale of 0 to 1) amongst categories.

The regions that impacted the classification in each category the most were widely varied across categories. For Local Private schools, an applicant being from California meant a higher likelihood of attending, while for a Local Public school, an applicant being from California meant a much lower likelihood of attending.

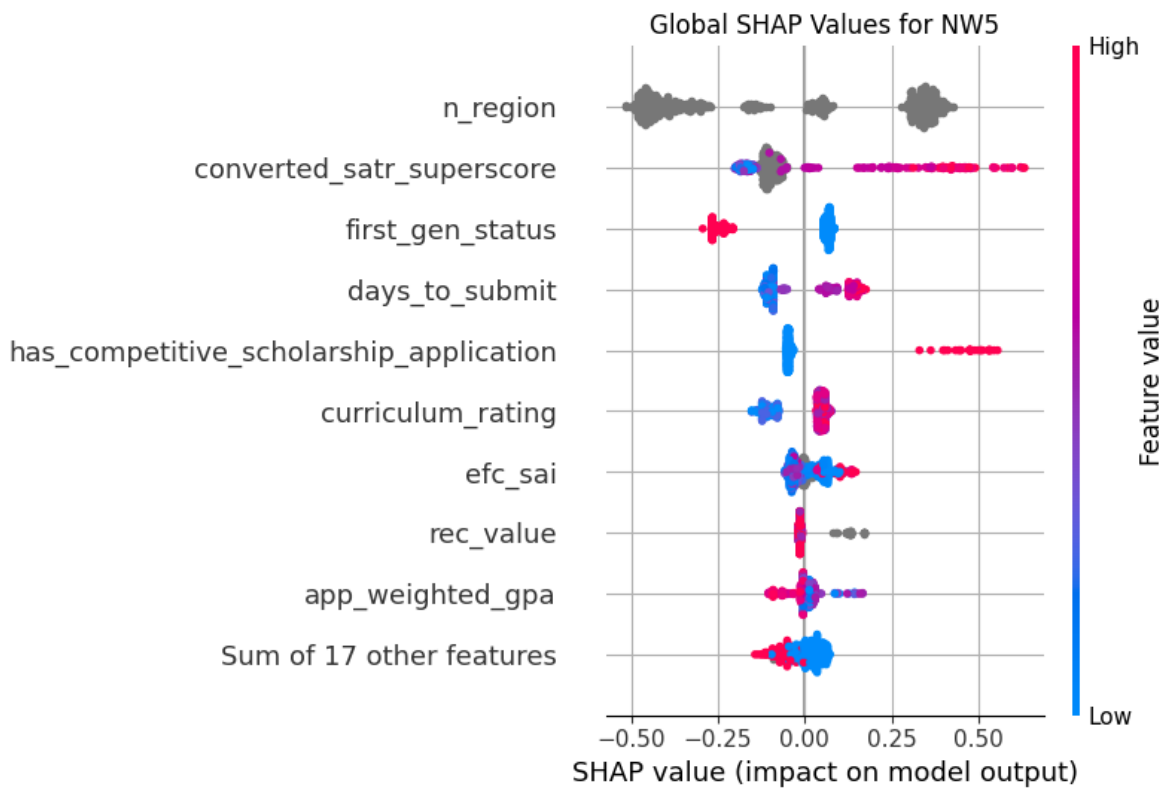


Figure 7: Graph of the importance and impact of features according to their values for Northwest 5 School Destination.

10 Discussion, Limitations, and Future Work

Interpretation: From the results, it can be seen that there are many factors that play a role in where a student enrolls. However, creating clusters and trying to sort students into a group will not result in a trustworthy prediction.

The factors that consistently show an impact on the final enrollment are an applicants' region of origin and whether or not they have visited campus. It is easy to speculate why these features can have a large impact. Visiting campus allows a student to visualize themselves attending that school. Regarding region, one could assume that students in the same general geographic region will remain there. It is important to note that any applicants who did not attend a school on the West Coast were not included in this data.

For Northwest 5 schools, the applicants most likely to attend are from California, other states, or the Bellingham Area of Washington. All Oregon regions indicate a smaller likelihood to attend a NW5 school, which may not be what a person would assume. Students with high standardized test scores and those who applied to a Willamette Competitive Scholarship are also more likely to attend a Northwest 5 School. This could indicate a propensity towards the liberal arts, which are encouraged and supported at Northwest 5 institutions.

A student from a secondary market state, California, the Greater Spokane Area of Washington, and the Bellingham area of Washington are more likely to attend local private schools. This could be due to the large number of private institutions in California and the many other private institutions in Washington and Oregon. Students who are athletes are more likely to attend a local private school. This makes sense due to the higher number of opportunities at private schools, with many of them competing at the Division 2 and Division 3 levels, which are less competitive to enter. If a student has a Division 1 offer, they are likely not looking at Willamette or other Division 3 schools.

The Three Large Public Schools in Oregon see the most enrollment from students residing in Oregon, as well as from non-student athletes, lower-income students, first generation students, and those with a higher GPA.

The region where a student is from is by far the most influential factor in determining if they will attend a Public school in Oregon, Washington, or California excluding the Three Large Oregon schools. Students in Washington, the Oregon Northern Valley (Coast) and Eastern Oregon are much more likely to attend these schools, with students from secondary states and California much less likely to attend.

Attendees of these public schools tend to submit applications faster and apply Regular Decision. This could indicate a later application date, quicker decision time, or both.

Limitations: This project was limited by the scale of the data that was accessed. Since Willamette does not receive as many applications as larger schools, the sample size was not very large. There is also a limitation in the timespan of data that can be accessed; only three years of applications were able to be used. Additionally, specific financial details were not accessed in this project due to complexity and PII concerns, which may limit the analysis of enrollment choices that are financially driven. Due to the nature of this project only using Willamette application data, the results and findings are limited to Willamette and are not generalizable.

Future work: More analysis could be completed on the interactions between features as they pertain to the enrollment destination. This would allow for more actionable results, as it would be easier to categorize students. The creation of an interactive model or function that would allow the Admissions and Enrollment team to insert a student's data and receive and predicted destination category would be extremely beneficial. To increase the robustness and accuracy of the model, there are a few directions that could be followed. Firstly, this work could be furthered with an increased sample size. At the time of the research, only the past three years of data were available. If this data is retained and new data is added, a more robust and accurate model could be created. Alternative modeling methods could also be utilized to work towards a more accurate model. Potential methods include Neural Networks and Naive Bayes.

11 Conclusions and Recommendations

There is no absolute, certain way to know what school an applicant will choose. However, using their region of origin, overall academic profile, and any financial information, one can gain a general understanding of where a student might enroll. For example, a student not from Oregon is not as likely to attend one of the 3 large Oregon public institutions.

Even if these predictions cannot be trusted to fully predict where a student will enroll, it can still provide some insight into how to attract them to Willamette. Since the models do not have an extremely high precision or recall rate, entire marketing and enrollment strategies should not be based off of these results.

12 Reproducibility and References

Reproduce main results:

```
cd your-repo
quarto render deliverables/M4-writeup-draft/writeup.qmd
```

Dependencies: The dependencies needed to reproduce this work can be found in the `requirements.txt` and `renv.lock` files in the root of the repository.

Data: URLs and approval status documented in `docs/data-dictionary.md`.

13 References

- [1] Anthony Schuette. 2023. Navigating the enrollment cliff in higher education. Retrieved from <https://files.eric.ed.gov/fulltext/ED628984.pdf>
- [2] Ahmad Slim, Don Hush, Tushar Ojha, and Terry Babbitt. 2018. Predicting student enrollment based on student and college characteristics. In *Proceedings of the 11th international conference on educational data mining*, 2018. International Conference on Educational Data Mining (EDM), Buffalo, New York, 383–389. Retrieved from https://educationaldatamining.org/files/conferences/EDM2018/papers/EDM2018_paper_136.pdf
- [3] Michael Soltys, Hung Dang, Ginger Reyes Reilly, and Katharine Soltys. 2021. Enrollment predictions with machine learning. *Strategic Enrollment Management Quarterly* 9, 2 (2021), 11–18.